

Duffing oscillator neural SDE

Analysis

Date revised
September 28, 2025

Moment matching turned **ON**

Moment matching ensures that—at every time step—the first (mean) and second (variance) moments of the neural SDE's distribution of trajectories matches the training data's distribution of trajectories.

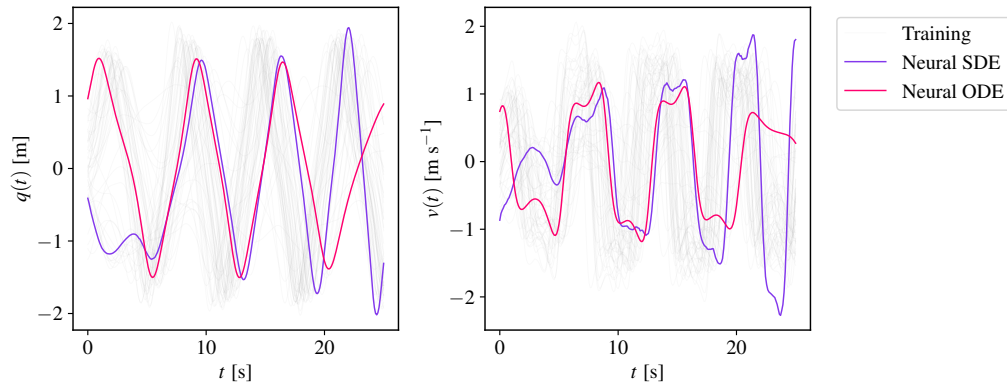


Fig. 1. Dynamics of the Duffing oscillator under the neural SDE and the neural ODE given a randomly sampled initial position and velocity, $q_0, v_0 \sim \mathcal{N}(0, 0.5)$.

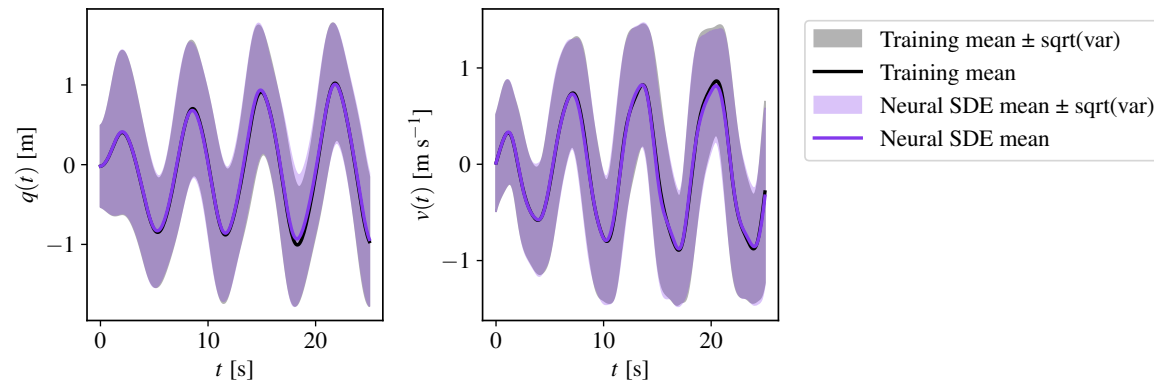


Fig. 2. Comparison of the distributions of neural SDE predictions and the model's training data, sampled from a range of initial conditions. The shaded regions are one standard deviation from the mean trajectory.

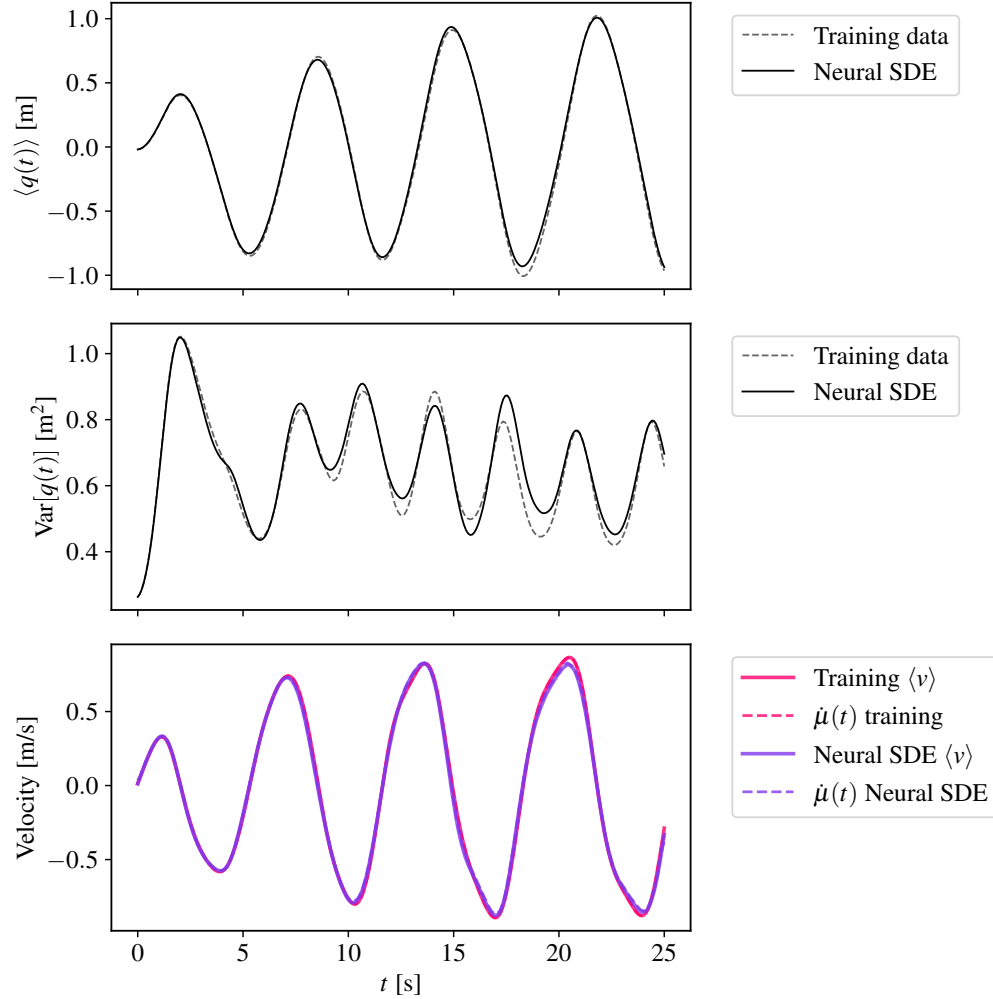


Fig. 3. **Top:** the evolution of the mean of the distribution of the trajectories. The training data (black, dashed) and the neural SDE predictions (black, solid) show good agreement throughout the domain of the problem, with some deviations observed later in the numerical integration. **Middle:** the evolution of the variance of the distribution of the trajectories, in the same style as the top plot. The variance has more visible deviation throughout the numerical integration. **Bottom:** the mean velocity of the training data (solid, pink), neural SDE (solid, purple), and the corresponding time derivatives of the first moment. The time derivative of the first moment of the distribution of trajectories should be the expected value of the velocity of the trajectories, $\dot{\mu}(t) = \mathbb{E}[v(t)]$ (by Kramers–Moyal coefficients).

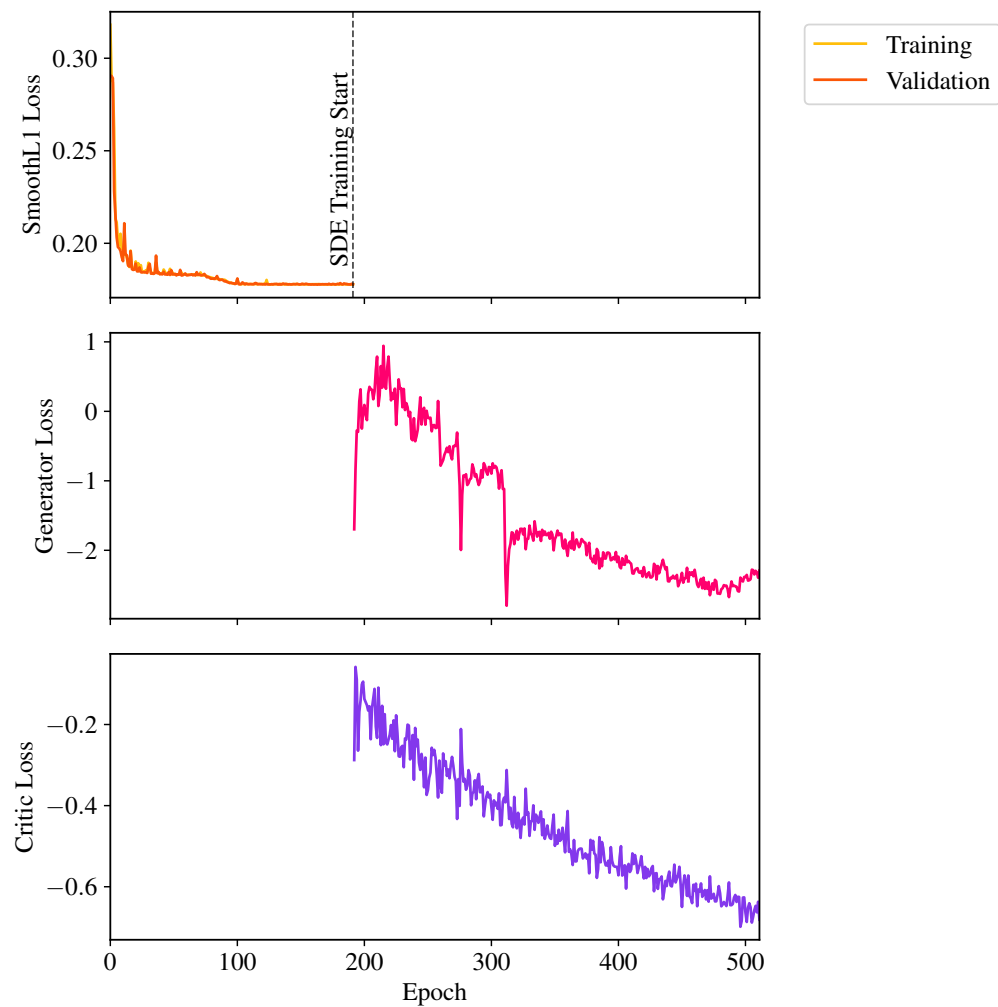


Fig. 4. **Top:** Smooth L1 Loss for the neural ODE training. **Middle:** generator loss for the neural SDE's generative adversarial loop. **Bottom:** critic loss for the neural SDE's generative adversarial loop.

Moment matching turned **OFF**

Moment matching ensures that—at every time step—the first (mean) and second (variance) moments of the neural SDE's distribution of trajectories matches the training data's distribution of trajectories. Here, we present results for the moment matching terms turned OFF in the cost function.

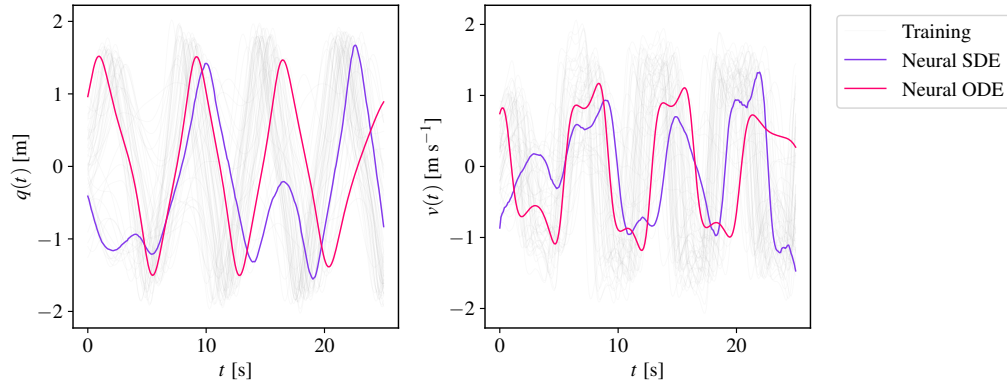


Fig. 5. Dynamics of the Duffing oscillator under the neural SDE and the neural ODE given a randomly sampled initial position and velocity, $q_0, v_0 \sim \mathcal{N}(0, 0.5)$.

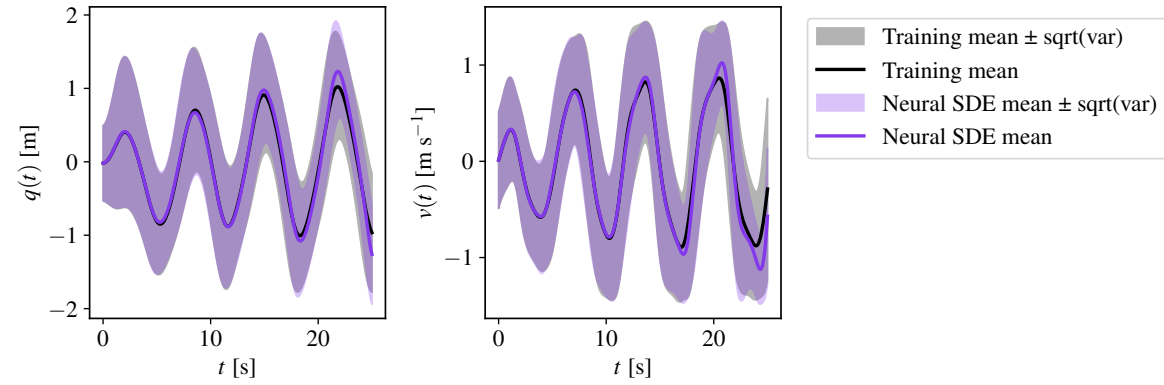


Fig. 6. Comparison of the distributions of neural SDE predictions and the model's training data, sampled from a range of initial conditions. The shaded regions are one standard deviation from the mean trajectory.

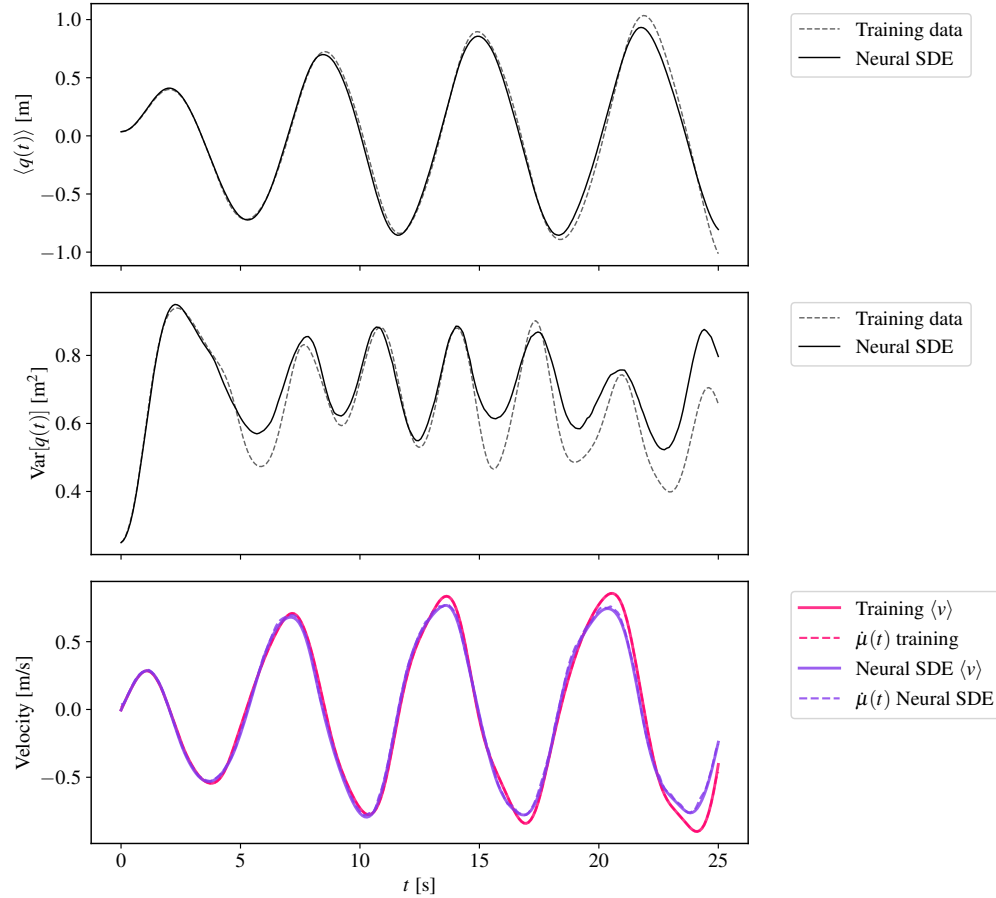


Fig. 7. **Top:** the evolution of the mean of the distribution of the trajectories. The training data (black, dashed) and the neural SDE predictions (black, solid) show good agreement throughout the domain of the problem, with some deviations observed later in the numerical integration. **Middle:** the evolution of the variance of the distribution of the trajectories, in the same style as the top plot. The variance has more visible deviation throughout the numerical integration. **Bottom:** the mean velocity of the training data (solid, purple), neural SDE (solid, pink), and the theoretical velocity (mint, dashed). The time derivative of the first moment of the distribution of trajectories should be the expected value of the velocity of the trajectories, $\dot{\mu}(t) = \mathbb{E}[v(t)]$.

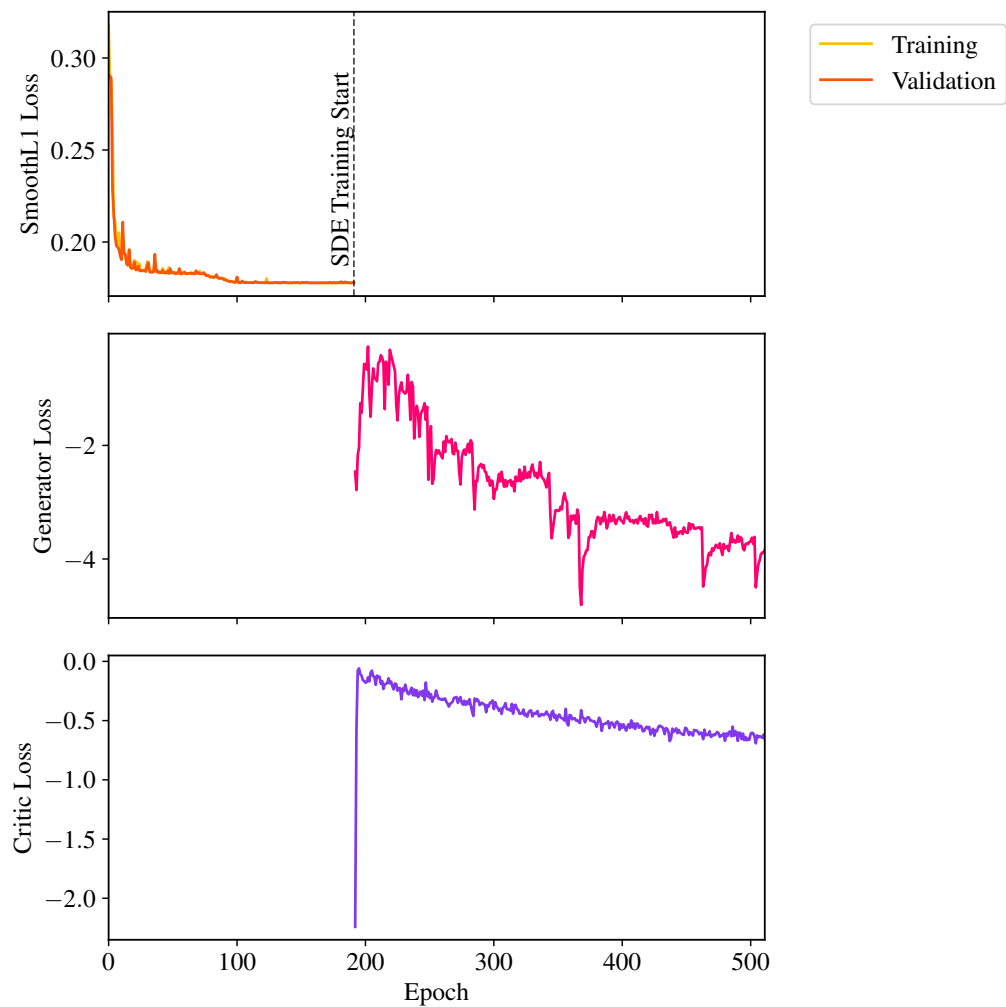


Fig. 8. **Top:** Smooth L1 Loss for the neural ODE training. **Middle:** generator loss for the neural SDE's generative adversarial loop. **Bottom:** critic loss for the neural SDE's generative adversarial loop.